BTECCE23605A:: Machine Learning

Course Type	PEC	Semester	7

Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course helps the student to identify the role of Supervised, Unsupervised and Reinforcement Algorithms. Illustrates concept learning and formulate research-based problems based on Machine Learning techniques. Analyzes different algorithms used in Machine Learning as applied to generic datasets.

Course Outcomes

CO No.	Statement
1	Demonstrate knowledge of Supervised, Unsupervised and Reinforcement algorithms through implementation for sustainable solutions of applications.
2	Evaluate different classification algorithms through implementation for sustainable solutions of applications.
3	Apply different clustering algorithms used in Machine Learning to generic datasets and specific multidisciplinary domains.
4	Analyze research-based problems using Neural Networks and genetic algorithms.
5	Evaluation of different convolution neural network algorithms on well formulated problems along with stating valid conclusions that the evaluation supports.

Mapping of COs to POs and PSOs

CO No			POs										PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1		3															UN
CO2				3													AP
CO3					2												AN
CO4							2										AP
CO5																3	AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction	Hours 9	CO	BTL
Introduction: What is Machine Learning? **Types of learning:** Supervised, Unsupervised, Reinforcement. **Simple regression**: Types, Making predictions, Cost function, Gradient descent, Training, Model evaluation. **Multivariable Regression:** Growing complexity, Normalization, Making predictions, Initialize weights, Cost function, Gradient descent, Simplifying with matrices, Bias term, Model evaluation **Learning System:** Well posed learning problems, Designing a Learning system, Applications of machine learning, Perspective and Issues in Machine Learning			1	UN
Unit No 2	**Classification Methods**	**Hours 9**	**2**	**AP**
Support Vector Machine Classification Algorithm, Hyper plane, optimal separating Hyper planes, kernel functions, kernel selection, Applications Introduction to Ensemble and its techniques, Bagging and Bootstrap Ensemble methods Introduction to Random Forest, growing of Random Forest, Random feature selection				
Unit No 3	**Clustering Methods**	**Hours 9**	**3**	**AP**
Overview of Clustering and Unsupervised Learning Introduction to Clustering methods: Partitioning methods K-Means algorithm, assessing quality and choose number of clusters, KNN (1 NN, K NN) techniques, K-Medians, Density based method: Density-Based Spatial Clustering Hierarchical clustering methods: Agglomerative Hierarchical clustering technique, Roles of Dendrograms and Choosing number clusters in Hierarchical clustering, Divisive clustering techniques, Principal Component Analysis (PCA)				
Unit No 4	**Neural network and Genetic Algorithms**	**Hours 9**	**4**	**AP**
Neural Network Representation – Problems – Perceptron – Multilayer Networks and Back Propagation **Algorithms** – Advanced Topics – Genetic Algorithms – Hypothesis Space Search – Genetic Programming – Models of Evaluation and Learning.				
Unit No 5	**Convolutional Neural Network**	**Hours 9**	**5**	**AN**
Convolutional Neural Network Recursive Neural Network, Recurrent Neural Network, Long-short Term Memory, Gradient descent optimization				

Textbooks

1	*T. Mitchell, "Machine Learning", McGraw-Hill, 1997.*
2	*Douglas Montgomery, Elizabeth A. Peck, and G. Geoffrey Vining, "Introduction to Linear Regression Analysis", 5th edition, Wiley publication*